Homework 1: Search Algorithms

Total: 100 marks Due: October 14, 2026

Important Regulations and Policies

Submission Guidelines

- Please submit your declaration and solutions on Blackboard (under the "course contents" tab).
- Your submission must be a **single PDF file**. Submissions in other formats (e.g., image files, Word documents) will not be graded.
- Please write your name and student ID on the first page of your solutions.
- For the late submission policy, please refer to regulations posted on Blackboard.

Collaboration Policy

- You are allowed to discuss the problems with your classmates to better understand the concepts.
- However, you **MUST** write up your own solutions independently.
- You **MUST NOT** copy from your classmates or from any other material. Do not share your final written solutions with others.

Policy on AI Tools (e.g., ChatGPT, Copilot, Gemini, Claude, Codex)

- Our policy is "Approach 2 – Use only with prior permission." For this assignment, you are permitted to use AI tools to help you understand concepts.
- **Acknowledgement is mandatory.** If you use an AI tool for any part of a problem, you **MUST** attach the full, unedited transcript of your interaction with the tool for that problem. This includes the prompts you used and the original answer generated by the AI. This should be attached at the end of your submission. Failure to do so will be considered a violation of academic integrity.

Problem 1: The Warehouse Delivery Cart **(15 marks)**

Imagine a cart moves on a static $M \times N$ grid (the map) to complete a delivery. The map and the location of a loading bay (the goal) are known. The cart must reach the loading bay and come to a complete stop, using as few actions as possible.

The cart follows these rules:

- The cart is directional and always faces one of four directions $d \in \{N, S, E, W\}$.
- Its velocity is an integer v satisfying $0 \leq v \leq V_{\max}$, where $V_{\max} \geq 1$. A velocity of 0 means that the cart is stationary.
- The cart can turn only when it is stationary. The actions `left` and `right` turn it by 90° in the corresponding direction without changing its position.
- The action `fast` first increases the velocity by 1, then moves the cart forward by a number of squares equal to the new velocity.
- The action `slow` first decreases the velocity by 1, then moves the cart forward by a number of squares equal to the new velocity.
- The cart cannot exceed the maximum velocity $V_{\max}$, have a negative velocity, pass through obstacles, or leave the grid. An action that would violate any of these restrictions is not allowed.
- Every action costs 1, regardless of how many squares the cart moves. The task is complete only when the cart is at the loading bay with velocity 0.

(a) (7 marks) What information must a search state record about the cart? Give a minimal sufficient state representation and determine the size of the state space. Justify your answer.

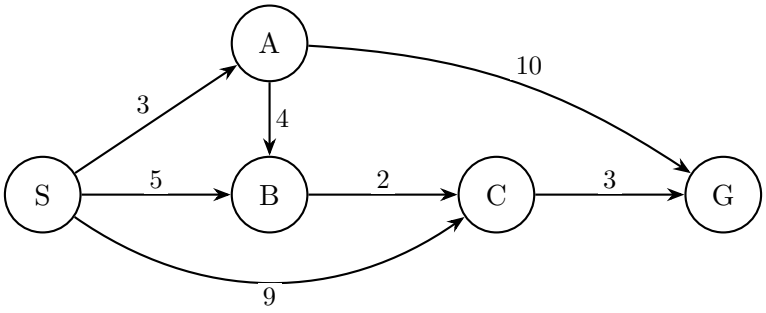
(b) (8 marks) Let the loading bay be at (x_g, y_g). Consider the straight-line distance heuristic

$$h(s) = \sqrt{(x - x_g)^2 + (y - y_g)^2}.$$

Is this heuristic admissible for every valid instance of the problem? Justify your answer.

Problem 2: Uninformed Search **(20 marks)**

Consider the state space graph shown below. S is the start state, and G is the goal state. The numbers on the edges represent the cost of traversing that edge.



For each of the following search algorithms, list the sequence of nodes expanded, the path returned, and its total cost. Assume that the successors of a node are generated in alphabetical order. In case of a tie in priority, expand the node that was added to the fringe first.

(a) (6 marks) Breadth-First Search (BFS).

(b) (6 marks) Depth-First Search (DFS).

(c) (8 marks) Uniform-Cost Search (UCS).

Problem 3: A* Search **(20 marks)**

Consider the same state space graph from Problem 2. The number on each edge $u \to v$ represents its traversal cost, denoted by $c(u, v)$. The path cost $g(n)$ is the sum of the edge costs along the current path from the start state to node n. We now introduce a heuristic function $h(n)$, which estimates the minimum remaining cost from node n to the goal. The heuristic values are given in the table below.

Node	S	A	B	C	G
h	9	8	5	3	0

(a) (15 marks) Perform A* search on this graph. Show the contents of the priority queue at each step of the algorithm. The priority queue should store nodes in the format $(\text{Node}, f(n))$, where $f(n) = g(n) + h(n)$.

(b) (5 marks) What is the solution path found by A* search, and what is its cost?

Problem 4: Heuristic Properties **(15 marks)**

Using the same graph, costs, and heuristic values from Problems 2 and 3:

(a) (7 marks) Is the heuristic h admissible? Justify your answer by comparing the heuristic value to the true optimal cost for each node.

(b) (8 marks) Is the heuristic $h(n)$ consistent? Justify your answer by checking the consistency condition for every edge in the graph. The consistency condition is $h(u) \leq c(u, v) + h(v)$ for an edge from node u to node v with cost $c(u, v)$.

Problem 5: Collecting Parcels **(15 marks)**

Imagine a robot is on a static $M \times N$ grid and must collect all K parcels. The map and parcel locations are known. Parcels occupy distinct, fixed squares. Entering a parcel's square automatically collects it. If a parcel is at the initial position, it is already collected in the initial state.

The robot can move one square up, down, left, or right. Each move costs **2 units of energy**; moves through obstacles or outside the grid are not allowed. Collecting a parcel has no additional cost, and the robot has no carrying-capacity limit. The objective is to collect all parcels using the minimum total energy. The task ends as soon as all parcels have been collected.

(a) (5 marks) Describe a minimal sufficient state representation and give the size of the state space. You do not need to exclude invalid or unreachable combinations. Justify your answer.

(b) For a state s, a heuristic $h(s)$ estimates the minimum additional energy needed to collect all remaining parcels. For each heuristic below, determine whether it is admissible and justify your answer. Both heuristics are defined to be 0 when no parcels remain.

(i) **(5 marks)** $h_1(s) = 2 \times r(s)$, where $r(s)$ is the number of remaining parcels.

(ii) **(5 marks)** $h_2(s) = 2 \times D(s)$, where $D(s)$ is the Manhattan distance from the robot's current position to the farthest remaining parcel.

Problem 6: Tree Search and Graph Search **(15 marks)**

(a) (4 marks) Explain the fundamental difference between tree search and graph search in their treatment of repeated states. What main problem does graph search aim to address?

(b) Consider the following claim:

> "An admissible heuristic guarantees that A* graph search finds an optimal solution, even if states in the closed set are never reopened."

A state is added to the closed set after it is expanded. Ignore any new path to a closed state. However, if a cheaper path to a state still in the frontier is found, update its path cost and parent. Stop when the goal is removed from the frontier.

(i) **(7 marks)** Construct a simple directed graph with no more than 4 nodes, positive edge costs, and an admissible but inconsistent heuristic that disproves the claim. Trace the frontier and closed set at each step. Compare the cost of the returned path with the true optimal cost.

(ii) **(4 marks)** Propose a specific modification that restores optimality when the heuristic is admissible but may be inconsistent. Briefly explain why it works. Assume a finite graph, positive edge costs, and a heuristic value of 0 at the goal.